

Flutter on macOS - Clean Teardown and Rebuild

Every code block below is pure commands. Nothing in them is a comment, a placeholder, or a note. Copy a whole block, paste it, press Enter, wait for it to finish, then move to the next one.

All explanation is in the plain text between the blocks. Never paste the plain text.

PART 1 - REMOVE EVERYTHING

1.1 Uninstall the Flutter SDK

```
brew uninstall --cask flutter
```

If that says the cask is not installed, you installed Flutter another way. Skip to 1.2.

1.2 Remove the development folder

```
rm -rf ~/development
```

1.3 Remove the Flutter and Dart caches

```
rm -rf ~/.pub-cache
```

```
rm -rf ~/.flutter
```

```
rm -rf ~/.dartServer
```

```
rm -rf ~/.flutter_tool_state
```

Some of these may not exist. `rm -rf` stays silent when that happens, which is fine.

1.4 Clean your shell profile

Open the file:

```
open -e ~/.zshrc
```

Delete any line that mentions `flutter`, `development/flutter`, or `Android/sdk`. Save the file and close it.

Then reload it:

```
source ~/.zshrc
```

1.5 Remove the Android SDK

Skip this section if you had Android Studio before this week and want to keep it.

```
rm -rf ~/Library/Android
```

```
brew uninstall --cask android-studio
```

```
rm -rf ~/Library/Application\ Support/Google/AndroidStudio*
```

```
rm -rf ~/Library/Preferences/AndroidStudio*
```

```
rm -rf ~/Library/Caches/Google/AndroidStudio*
```

```
rm -rf ~/Library/Logs/Google/AndroidStudio*
```

1.6 Remove the Gradle cache

This is several gigabytes. Removing it means the first build after reinstalling downloads again.

```
rm -rf ~/.gradle
```

1.7 Confirm it is gone

```
which flutter
```

```
which dart
```

```
which adb
```

All three should print nothing. If any still prints a path, close the terminal, open a new one, and run them again.

PART 2 - CLEAN INSTALL

Run these in order. Wait for each to finish before starting the next.

2.1 Install Flutter

```
brew install --cask flutter
```

This downloads over a gigabyte. It takes several minutes.

2.2 Check it

```
flutter --version
```

You should see a version number and `Channel stable`. Write that version down.

If it says `command not found`, run this and then try again:

```
echo 'export PATH="/opt/homebrew/bin:$PATH"' >> ~/.zshrc
```

```
source ~/.zshrc
```

2.3 Install Android Studio

```
brew install --cask android-studio
```

2.4 Run the Android Studio setup wizard

Open Android Studio from Applications. Choose **Standard** install when the wizard asks. Let it finish downloading. Close Android Studio when it reaches the welcome screen.

2.5 Install the command-line tools

This is the step everybody misses.

Open Android Studio. On the welcome screen click **More Actions** and then **SDK Manager**.

In the left panel click **Android SDK**. Three tabs appear on the right: **SDK Platforms**, **SDK Tools**, **SDK Update Sites**.

Click the **SDK Tools** tab and tick these four:

- Android SDK Build-Tools
- Android SDK Command-line Tools (latest)
- Android SDK Platform-Tools
- Android Emulator

Click **Apply**, accept the licence dialogue, wait for the download, then click **OK**.

If you cannot find it: **Android Studio** → **Settings (⚙)** → **Languages & Frameworks** → **Android SDK** → **SDK Tools**.

2.6 Check again

```
flutter doctor
```

2.7 About the Android licence warning

You will probably see this:

```
Android license status unknown.
```

And if you run `flutter doctor --android-licenses` it replies that the option is no longer needed.

This is not your machine and not a mistake. Android replaced the old `sdkmanager` tool with a new `android` CLI, and Flutter's licence check has not caught up yet. It is an open Flutter bug. Your licences are accepted and your builds will work.

Confirm the licence files exist:

```
ls ~/Library/Android/sdk/licenses
```

If files are listed, you are licensed. Move on.

2.8 The test that actually matters

`flutter doctor` can be wrong. A build cannot.

```
cd ~
```

```
flutter create sanity_check
```

```
cd sanity_check
```

```
flutter build apk --debug
```

Look for a line saying it built `app-debug.apk`. If you see it, your setup is correct no matter what doctor says about licences.

2.9 Delete the test app

```
cd ~
```

```
rm -rf ~/sanity_check
```

PART 3 - CREATE AN EMULATOR

Open Android Studio. On the welcome screen click **More Actions** and then **Virtual Device Manager**.

Click **Create Device**. Choose **Pixel 8**. Click **Next**.

Choose the latest **API 35** system image. If it has a download arrow next to it, click the arrow and wait.

Click **Next**, then **Finish**.

Click the ▶ play button to start it. The first boot takes one to three minutes.

Then check Flutter can see it:

```
flutter devices
```

PART 4 - VS CODE

Install VS Code if you do not have it:

```
brew install --cask visual-studio-code
```

Open VS Code. Press X to open Extensions. Search for **Flutter** and install it. It installs the Dart extension automatically.

PART 5 - OPTIONAL: XCODE FOR iOS

Only needed if you want to build for iPhone. It is a 10 GB download. Install Xcode from the App Store first, then:

```
sudo xcode-select --switch /Applications/Xcode.app/Contents/Developer
```

```
sudo xcodebuild -runFirstLaunch
```

```
brew install cocoapods
```

Then:

```
flutter doctor
```

PART 6 - CREATE THE COURSE PROJECT

```
cd ~
```

```
mkdir -p ~/dev
```

```
cd ~/dev
```

```
flutter create --org za.co.moint quote_app
```

```
cd quote_app
```

```
flutter run
```

The first build takes three to eight minutes. Every build after that takes seconds.

While it is running, in that same terminal:

Press `r` to hot reload. Press `R` to hot restart. Press `q` to quit.

QUICK REFERENCE

Check the install:

```
flutter doctor -v
```

Update Flutter:

```
flutter upgrade
```

Clear a broken build:

```
flutter clean
```

```
flutter pub get
```

Find where things are installed:

```
which flutter
```

```
echo $HOME/Library/Android/sdk
```

Remove a duplicate adb if doctor complains about multiple binaries:

```
brew uninstall --cask android-platform-tools
```

IF SOMETHING GOES WRONG

What you see	What to do
<code>command not found: flutter</code>	Close the terminal, open a new one. If still missing, redo 2.2
<code>cmdline-tools component is missing</code>	Redo step 2.5
<code>Android license status unknown</code>	Expected. See 2.7. Prove it with 2.8
<code>Multiple adb binaries found</code>	<code>brew uninstall --cask android-platform-tools</code>
Emulator will not start	Restart the Mac, then start it again from Virtual Device Manager
Build fails for no clear reason	<code>flutter clean</code> then <code>flutter pub get</code> then build again
<code>zsh: command not found: #</code>	You pasted an explanation line. Only paste the code blocks
<code>zsh: no matches found</code>	You pasted a line with a <code>*</code> in it that matched no file